



Probing the interaction of carbonaceous dots with transferrin and albumin: Impact on the protein structure and non-synergetic metal release

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ABSTRACT

Carbon quantum dots (CQD) have been emerging as an essential material for biological applications due to their relevant properties. Here, we synthesized and characterized low toxic anionic carbon dots and examined the interaction of the nanoparticles with human transferrin (hTf) and bovine serum albumin (BSA) by using multi-spectroscopic techniques. CQD showed low cytotoxic effect against HBMEC, A549, and HTC116 cells. Temperature variable fluorescence experiments reveal that CQD quench the BSA and hTf intrinsic fluorescence following a dynamical and static mechanisms, respectively. Both proteins attach tightly to the nanoparticle ($K_a \approx 10^4$ – 10^5 M^{-1}), and hydrophobic forces drive the binding. Additionally, hTf-CQD interaction is also guided by electrostatic interaction. Furthermore, protein CD spectra reveal that both proteins experience structural changes during nanoparticle interaction, which differs from previously reported anionic carbon dots from other sources. Experiments corroborate the observed structural modifications and unveil that the nanoparticle interaction induces iron release from hTf lobes. The mechanism of hTf iron release induced by CQD can be understood based on anionic non-synergetic effects.

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1. Introduction

Carbonaceous quantum dots (CQD) are a new kind of carbon nanomaterials obtained during the purification of single-wall nanotubes [1]. These dots gradually became a star in the nano-carbon family, due to their functional, abundant and economic relevance. It also presents several advantages like their low toxicity, water-solubility, superior physicochemical stability, excellent biocompatibility, good cell permeability, high photostability, and higher sensitivity when compared with other nanomaterials. Additionally, the CQD are also promising nanomaterials for various applications in catalysis, electrocatalysis, and photocatalysis, sensors, biosensor, biomedicine, bioimaging, optoelectronics, drug delivery and photodynamic therapy [2]. Despite various potential biological applications, previous studies involving CQDs have not observed functional effect over plasma proteins like albumin [3] and transferrin [4].

Serum albumin is a single-chain protein (~67 kDa) and composed of three homologous domains, each of them comprising two subdomains.

The protein is highly abundant in the circulatory system, which includes around 60% of total plasma protein. Although albumin has other relevant functions, the most notable one is its capability to attach reversibly and transport many hydrophobic and negatively charged small molecules [5]. Besides, Human serum albumin is also the main component of protein corona [6,7].

Transferrin is a bilobal protein (termed N- and C-lobes) and divided into four subdomains (N1, N2, C1, and C2). One Fe^{+3} ion binds tightly and reversibly to each lobe through coordination to identical protein residues: two tyrosine residues, one aspartic acid, and one histidine. Besides, carbonate anion is attached to the conserved arginine residue and act as synergistic anion to the distorted octahedral coordination. hTf travels in the blood as di-ferric (Fe_2hTf), mono-ferric N-lobe hTf (FeNhTf), mono-ferric C-lobe hTf (FeChTf) and iron-free hTf (apohTf) [8]. Due to other relevant second shell residues [8,9], the two lobes respond distinctly to pH, anions, and the conformation of the other lobe.

Biochemical properties of individual proteins and their intrinsic structural flexibility play an essential role in controlling surface-driven modifications to the protein secondary structures [10]. In regard to these role, many studies explore key structural and biochemical features of binary or even ternary complexes, which are resulting from the plas-matic [11–16] and non-plas-matic protein [17] interactions with drug

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molecules. Besides protein properties, the morphology of the nanoparticles may also contribute to protein structural changes. When compared to planar, curved NP surfaces offer extraordinary adaptability and extra surface area to attached protein molecules [18], which can, in some cases, disturb its secondary structure [19] irreversibly. Countless studies in the literature rely on physicochemical aspects of protein interactions with many nanomaterials [8–10]. By one side, changes in the Bovine Serum Albumin (BSA) secondary structure induced by Au-NPs [19–21], ZnO-NPs [22] has been reported. By the other side, no significant conformational changes were observed when BSA was absorbed in C60 fullerene NP [23] surface. Furthermore, tubulin exhibited conformational change and reduction in the polymerization, which was influenced by TiO₂-NPs [23–25]. Transferrin undergoes irreversible conformational change by interaction with SPIONs [9].

A549 cells are well known to be more sensitive to toxic iron effects [26], while HTC116 cells are more resistant to those effects [27]. Several mechanisms have been reported for nanoparticles cytotoxicity, which includes ROS production, oxidative stress, lysosomal damage, dissolution and release of toxic metals, protein aggregation and fibrillation, inflammation, disruption of protein conformation [28]. Furthermore, models of intestinal tissues (in vivo and in vitro) showed that chronic and acute oral exposure to low toxic polystyrene nanoparticles might affect iron uptake and iron transportation [29]. Due to cell membrane disruption caused by nanoparticles, iron transportation have increased during an exposition of high doses of nanoparticles and chickens in contact with 50 nm carboxylated nanoparticles present lower iron absorption, which was compensated by intestinal villi remodeling, than control animals.

In order to explore the subtle effects of low toxic nanoparticles at a molecular level, the present investigation applied biophysical techniques like UV-Vis, temperature variable spectrofluorimetry, circular dichroism and MTT assays to elucidate the functional consequences (structural changes and metal release) over BSA and hTf proteins induced by anionic low toxic CQD interaction, as well as their impact on the Brain Microvascular Endothelial (HBMEC), human lung adenocarcinoma (A549) and human colon cancer (HTC116) cell culture media.

2. Material and methods

2.1. Materials

Bovine Serum Albumin (BSA) and human Transferrin (hTf), which has 300–600 ppm iron content, were purchased from Sigma Aldrich (USA). Cdots (5 nm, were fabricated by pyrolysis and the size were determined by MET). The stock solutions of proteins were prepared in 10 mM tris (hydroxymethyl) aminomethane (Tris-HCl; pH 7.5) and sodium phosphate buffer (pH = 7.5).

2.2. Synthesis of carbon dots

The carbon nanoparticles were obtained by pyrolyzing 30 mg of Concanavalin A (type VI, Sigma). The protein concanavalin A was placed in a furnace at 300 °C for 3 h and cooled to room temperature. Then 3 mL of 70% ethanol was added to extract the nanoparticles. The mixture was sonicated in Ultrasonic bath (LimpSonic) for 30 min. The resulting solutions were centrifuged twice at 15,000 rpm for 30 min, and the supernatant was extracted. For purification purposes, the nanoparticles were left in an oven at 100 °C and later the carbon dots were removed in water. The samples were placed in Amicon Ultra-0.5 10 K (Millipore) ultrafiltration devices with a volumetric capacity of 500 µL. The filters were placed into collector tubes and centrifuged at 12,000 rpm for 10 min at 4 °C, yielding 500 µL of concentrated carbon dot samples at each centrifugation.

2.3. MTT assay

HBMEC, A549, and HCT-116 cells were pre-cultured and plated in 96-well plates (20,000 cells/well), and after 24 h, the cells were incubated in an oven at 37 °C and under an atmosphere of 5% CO₂ for 24 h with the carbon dots at the following concentrations: 0.05, 0.125, 0.25, 0.5, 1 and 2 mg/mL or with culture medium alone. After 20 h of treatment, 20 µL of MTT (2.5 mg/mL) was added to the wells. The plate was further incubated for 4 h for MTT metabolism. Subsequently, the medium was discarded, and the crystals formed in the cells were solubilized in 100 µL of DMSO. Absorbance reading was performed on a plate reader (Molecular Devices - SpectraMax Microplate Reader) at 570 nm and 650 nm. The toxicity of the carbon dots was calculated as a function of the control, considered as 100% cell viability.

2.4. Transmission electron microscopy

In the analysis by transmission electron microscopy, the images were constructed by high power electron beam diffraction, incident on the surface of the sample, 5–12 nm thick, through the microscope Tecnai Spirit TEM, National Center for Structural Biology and Bioimaging (CENABIO 3) of the Federal University of Rio de Janeiro; operating at 120 kV. Samples were prepared by dropping the diluted nanoparticle suspensions into 200 mesh carbon covered copper grids with immediate evaporation of the excess solvent. Also, Energy Dispersive Spectrometry (EDS) was used for the identification and determination of the composition of the elements present in the samples investigated.

2.5. Fluorescence quenching

All protein quenching experiments were performed on a Cary Eclipse fluorescence spectrometer (Varian, Sydney, Australia). Slit widths of 10 nm were used for both excitation and emission, and temperatures were set to 293, 298 and 303 K. The protein solution and carbon dot-protein complexes were all equilibrated before the experiment. Measurements were performed using 3 µM protein concentration for BSA and hTf in buffer solution tris pH = 7.5, with excitation wavelength at 280 nm and emission spectra were recorded between 300 and 430 nm. The concentration range of carbon dots was 0 to 1.6 µM. The data were estimated as an average of three parallel experiments. The observed fluorescence intensities were corrected for the absorption of the excitation light and re-absorption of emitted light (inner filter effect [30]). The Stern-Volmer constant (K_{sv}) was determined by the intensity ratio

$$\frac{F_0}{F} = 1 + K_{sv}[Q] \quad (1)$$

where F₀, F are respectively the protein fluorescence intensity in the absence and presence of the quencher, [Q] is the quencher concentration. Also, the binding constants K_a were determined using the equation

$$\log\left(\frac{F_0 - F}{F}\right) = \log\log K_a + n \log[Q] \quad (2)$$

Also, the thermodynamic free energy changes were calculated by $\Delta G = -RT \ln K_a$. ΔH and ΔS were obtained by fitting the K_a values to the Van't Hoff equation

$$RT \ln K_a = \left(-\frac{\Delta H}{R}\right) \frac{1}{T} + \frac{\Delta S}{R} \quad (3)$$

K_a provides information about the affinity of protein to the CQDs while K_{sv} is related to quenching mechanism involved in the interaction.

2.6. UV-Visible absorption measurements

Spectra were recorded on a UV-Vis-NIR spectrophotometer (Erkin Elmer-lambda 650) using a cuvette with of 1 cm path length. The spectra were recorded between 400 and 500 nm. The concentration of hTf was 3 μ M in sodium phosphate buffer pH = 7.5 and Cdots the concentration series of carbon dots was 0 to 4.8 μ M.

2.7. Circular dichroism experiments

Circular Dichroism analyses were carried out at increasing CD concentrations. The concentration range of carbon dots was 0 to 1.6 μ M (5 nm), and the concentration of BSA and hTf were 6 μ M in sodium phosphate buffer pH = 7.5. Spectra were obtained at each BSA-Cdots

and hTf-Cdots concentration in a Chirascan spectropolarimeter (Applied Photophysics, London, UK) at 25 °C, using a quartz cuvette with 0.01 cm path length. Final spectra correspond to the average of three scans obtained from 190 nm to 260 nm wavelengths at a 25 nm/min speed after subtracting the spectra baseline corresponding to the buffer.

3. Results and discussions

3.1. Nanoparticle synthesis and characterization

Here, CQDs have been synthesized and characterized. Transmission electron micrograph (Fig. 1a) reveals that the carbon dots are homogeneous distributed, composed primarily of 76% carbon and 15% oxygen (Fig. 1b) and with an average size of $\approx$ 5 nm (Fig. 1c). Additionally, the

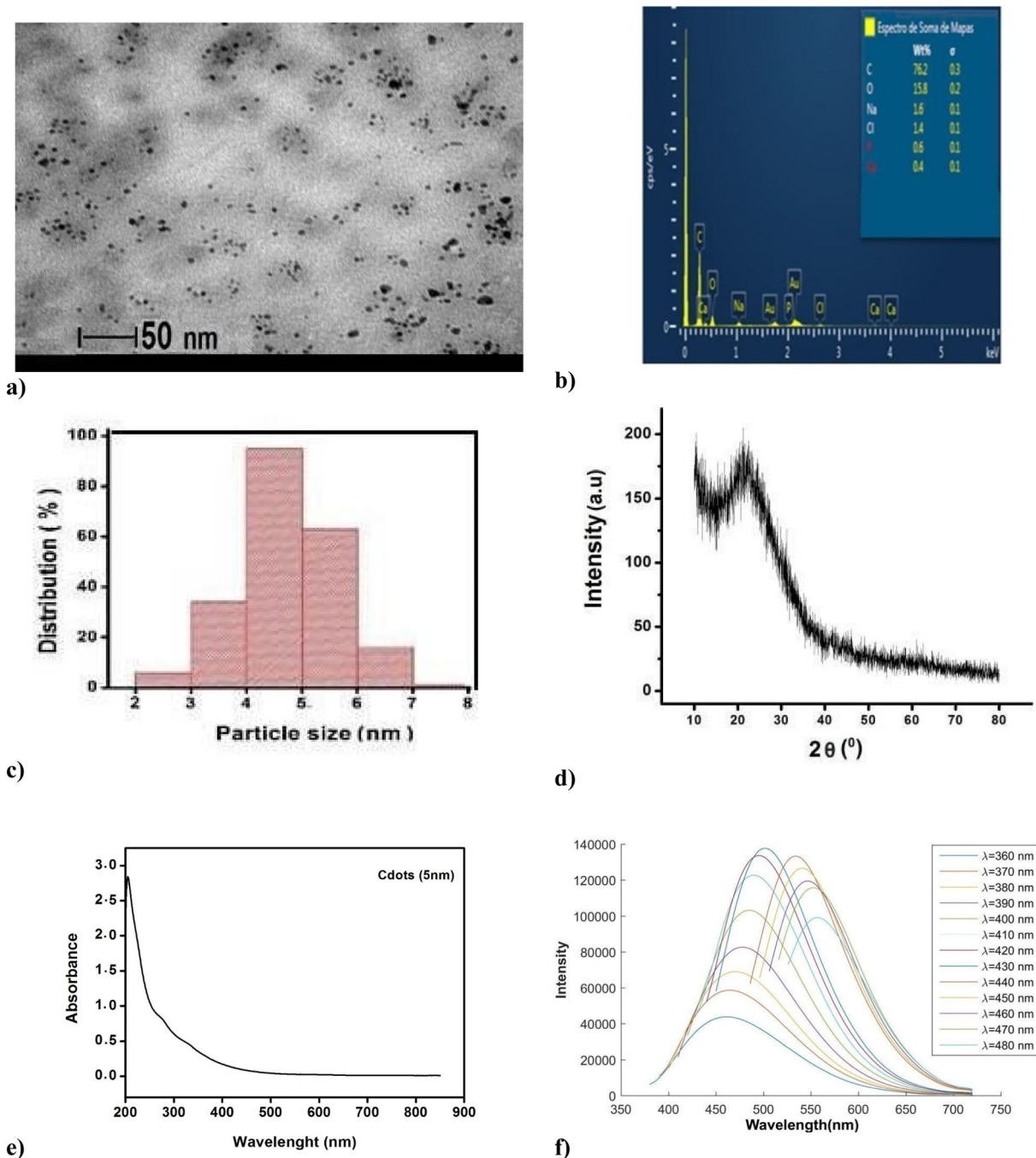


Fig. 1. Characterization of the carbon dots: (a) Transmission electron microscopy, (b) EDS showing the composition of carbon dots, (c) size distribution, (d) the PXRD pattern, (e) UV-Vis spectrum and (f) Fluorescence emission profile of the nanoparticles.

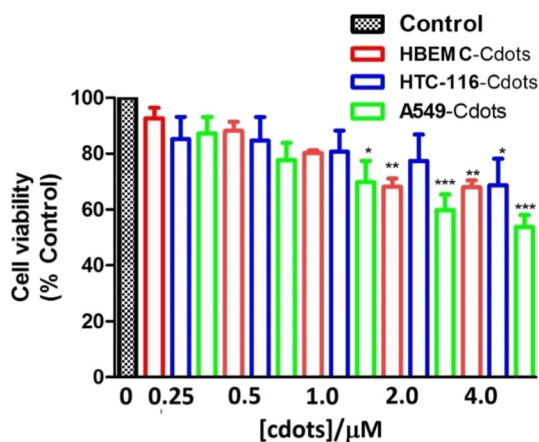


Fig. 2. Cytotoxicity of the carbon dots against the Human Brain Microvascular Endothelial cell (HBMEC), adenocarcinomic human alveolar basal epithelial cell (A549) and human colon cancer cell (HTC116) lines.

nanoparticles show low crystallinity (Fig. 1d), UV-Vis absorption in the range of 200–500 nm (Fig. 1e) and a typical fluorescence profile (Fig. 1f), where the emission spectra are red-shifted and change intensities as the absorption wavelength increases [31,32]. Also, measured zeta potential (-28.2 ± 1.8 mV) revealed that the synthesized carbon nanoparticles are anionic.

3.2. Cytotoxic assays

The development of in vitro assays to evaluate the cytotoxic potential of carbon dots represents a challenge due to the fast changes of its intrinsic physical-chemical properties [33] in the biological fluids dispersions that can strongly affect the biological outcomes tests for evaluating toxicity in biological systems [34,35]. In this conjecture, valuable efforts are required to develop methods to decrease the harmful and toxic effects of these nanomaterials in living organisms. As already mentioned, carbon-dots exhibit low toxicity against many cell lines, with some residual toxicity only for high nanoparticle concentration. Given that, MTT assay was performed for investigating carbon dot cytotoxicity by incubating the nanoparticles with Human Brain Microvascular Endothelial (HBMEC), human lung adenocarcinoma (A549) and human colon cancer (HTC116) cell lines. The results are presented in Fig. 2. As can be seen, carbon dots exhibit no representative cytotoxic effect for the three cell lines until 1.0 μM concentration, confirming the low toxic aspects of the nanoparticles synthesized here.

3.3. Interaction studies

Aromatic residues are the main contributors to the intrinsic protein fluorescence, making those residues a sensitive fluorescent probe to

investigate protein interactions with a guest molecule or nanomaterial. Fig. 3a shows the quenching of hTf during titration with the carbon dots at 298 K and where can be observed the presence of an isobestic point indicating the existence of free and bound CQDs (in equilibrium). By fitting the fluorescence intensity ratio F_0/F for different quencher concentration $[Q]$ to Eq. (1), it allows determining the Stern-Volmer constant K_{sv} . Fig. 3b shows the fitting results, where we can observe a linear behavior of intensity ratio F_0/F for the temperatures 293, 298 and 303 K. Besides, the determined K_{sv} increases as the temperature increases, suggesting that the hTf quenching process is guided by a dynamic mechanism [36].

Apart from the quenching mechanism, circular dichroism spectroscopy was used to monitor structural changes during the interaction of carbon nanoparticles with hTf (Fig. 4a) and BSA (Fig. 4b). Dichroic spectra of both proteins exhibit two negative peaks at 208 and 222 nm, representing the $\pi-\pi^*$ and $n-\pi^*$ electronic transitions in α -helix structure, respectively. As carbon dots have been titrated to the protein solutions, the two negative peaks of BSA and hTf reduce the intensity, meaning that both proteins undergo structural changes due to nanoparticle interaction by losing α -helix contents. Alterations in the HSA structure and function are mainly accomplished by saturable binding of some drugs. Similarly, modification of the BSA secondary structure revealed in Fig. 4a may be related to the hydrophobic interaction and hydrogen bonding established in the CQD-HSA complex. As consequence, β -sheet structure in the BSA becomes more relaxed and then exposing certain hidden groups in the protein.

Furthermore, the affinity of carbon nanoparticles to the proteins are measured using a modified Stern-Volmer equation (Eq. (2)). Fig. 4c shows the experimental data fitting. The obtained $K_a \approx 10^4-10^5 \text{ M}^{-1}$ imply that hTf bind firmly to the carbon dots and the increase on the temperature lead to more stable complex CQD-hTf, according to K_a dependence on the temperature. Additionally, Table 1 reveals that binding site number decreases (from $n = 1.10$ to 0.97) as the temperature increases (from $T = 293$ to 303 K) for the interaction of the CQDs and hTf. Such temperature behavior of n is related to the fact that the molecules become more disordered and experiences fast vibration and diffusion at higher temperatures, which in turn may lead to instability of CQD-protein complex.

3.4. Thermodynamical aspects of the interaction

Protein binding site and thermodynamic attributes of the interaction are also relevant features of the bio-interface [37]. The thermodynamic properties ΔH and ΔS for CQD-hTf interaction were determined (Fig. 4d) using Eq. (3). It can be noted that the determined $\Delta H = +35.4 \text{ kJ} \cdot \text{mol}^{-1}$ and $\Delta S = +210 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ are positive. Since both thermodynamic properties are positive for the interaction of the nanoparticle with hTf, it indicates that hydrophobic interaction as the main driving force [38,39]. However, due to the high values of ΔH , it does

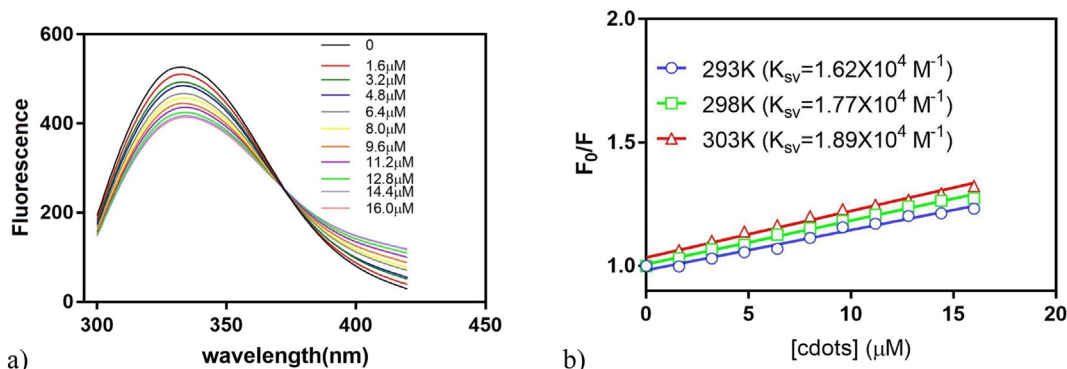


Fig. 3. Quenching the intrinsic fluorescence of hTf at 298 K by carbon dots (0–16 μM) by nanoparticle titration (c) and (b) individual Stern-Volmer plots.

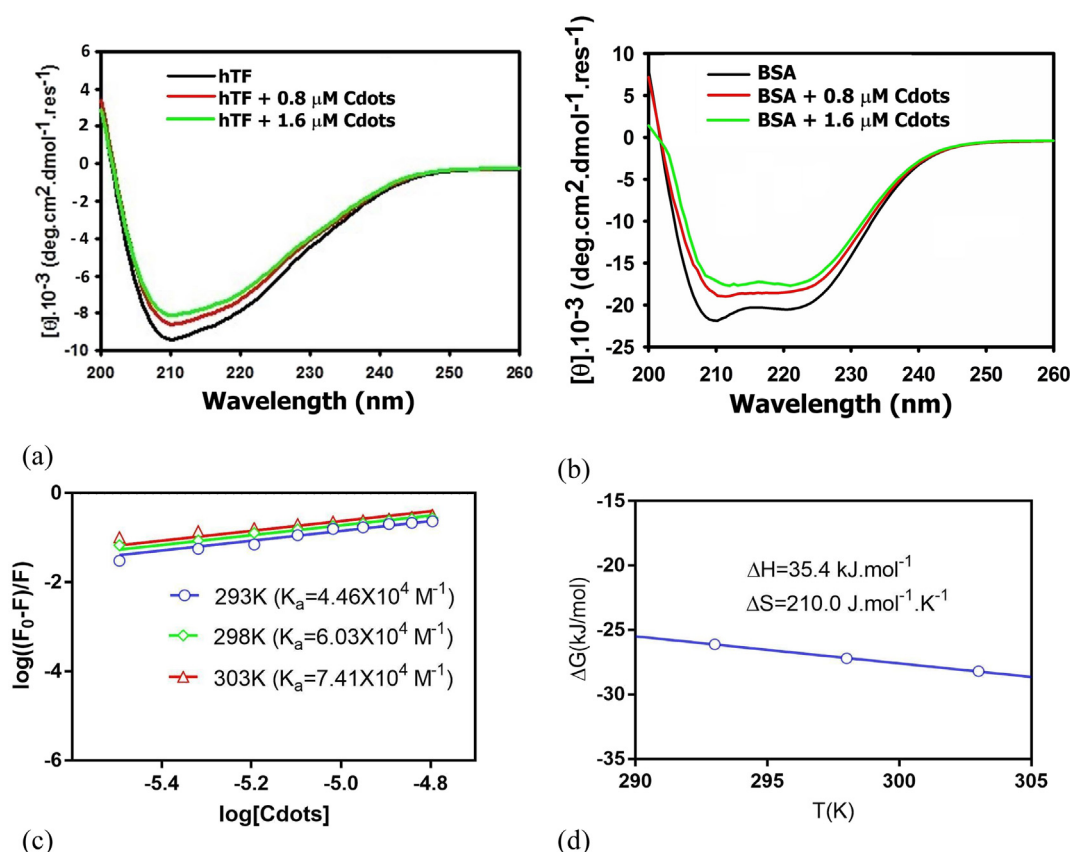


Fig. 4. Structural changes monitored by the circular dichroism spectra of (a) hTf and (b) BSA during nanoparticle titration. Modified Stern-Volmer plots (c) for the determination of K_a and (d) Van't Hoff plots for the determination of the thermodynamical parameter of the hTf-CD interaction.

not eliminate the participation of electrostatic interaction (resulting in negative chemical groups in the carbon dot surface and the hTf irons).

For comparison, Table 1 compiles the interaction and thermodynamic parameter obtained for the CQD-hTf together with CQD-BSA complex. As can be seen in Table 1, carbon dots quenches the BSA fluorescence following a static mechanism (K_{sv} values decreases as the temperature increases), which differs from the dynamic mechanism presented by hTf. BSA shares very comparable affinities to carbon dots as hTf. Nevertheless, due to the temperature dependence of the K_a , interaction BSA carbon dot is 3.8 times less endothermic and 1.7 times less entropy than hTf-carbon dot. Similar reported carbon dots [4] identified hTf iron (Fe^{3+}) as the binding site for the anionic carbon nanoparticle and due to the determined changes of enthalpy ($\Delta H = -1.81 \text{ kJ.mol}^{-1}$) and entropy ($\Delta S = 78.89 \text{ kJ.mol}^{-1}\text{.K}^{-1}$), it was identified electrostatic interaction as the principal promoter of the complex carbon dot- hTf stability. In this context, the nanoparticle obtained here reveals an intricate pattern of forces components in opposite to the single force previously reported [4].

Albumin structural modifications induced by CQD interaction and the high determined affinities to the protein may interfere in the in vivo protein functions such as blood reservoir and transport of endogenous and exogenous molecules. In particular, pharmacokinetic and pharmacodynamic properties are highly dependent on the drug

albumin binding affinities [40–42]. Nevertheless, in opposite to drug case exhaustively in vivo studies have still to be done to fully elucidate many aspects of the nanoparticle metabolism, distribution, excretion and interaction with tissues.

3.5. hTf iron release induced by carbon nanoparticles

Any functional alteration in the structure has a direct impact on the biological roles played by the proteins. Particular attention should be dedicated to hTf. Although hTf iron is naturally released by pH changes under physiological conditions [8], protein alteration induced by nanoparticles can also unleash premature iron release from its lobes and initiating toxic iron overload over cells [43], which creates challenges to use low toxic nanoparticle as nanomedicine.

UV-Vis spectroscopy has been used to evaluate the binding from the hTf iron point of view [8]. The characteristic peak ($\lambda_{\text{max}} = 475 \text{ nm}$) of the hTf iron was monitored as the nanoparticles have been titrated to the protein solution. Fig. 5 displays a decrease in the characteristic peak intensity by the carbon dots interaction. The reduction in the characteristic peak intensity in the UV-Vis during carbon dot titration (Fig. 5) indicates a change in the hTf iron chemical environment and suggests that Fe^{+3} has been released from the hTf.

Table 1

Thermodynamic parameters for hTf and BSA (in square brackets) determined from fluorescence quenching experiments.

T	n	K_{sv} ($\times 10^4 \text{ M}$)	K_a ($\times 10^4 \text{ M}$)	ΔH (kJ.mol^{-1})	ΔS ($\text{J.mol}^{-1}\text{.K}^{-1}$)	ΔG (kJ.mol^{-1})
293 K	1.10[0.99]	1.62[1.21]	4.46[6.41]	+35.4[9.34]	+210.0[123.8]	−26.1[−26.9]
298 K	0.99[1.01]	1.77[1.07]	6.03[6.87]	–	–	−27.2[−27.5]
303 K	0.97[0.98]	1.89[0.95]	7.41[7.28]	–	–	−28.2[−28.1]

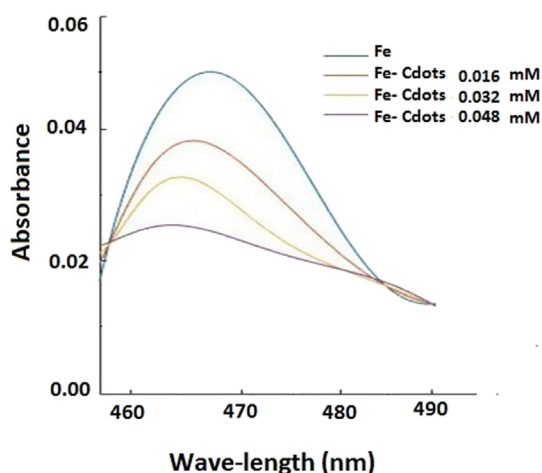


Fig. 5. UV-Vis region relative to hTf iron during protein titration.

In contrast to the structural changes observed here (Fig. 3a–b), reports involving the interaction of anionic carbon dots from other sources with BSA [3] and hTf [4], did not observe any structural change induced by nanoparticle interaction. These differences can be understood based on the synergetic anionic binding. To accomplish the well-known transferrin high affinity of Fe^{3+} , synergetic anionic binding to hTf has to co-occur. Arginine residue in each lobe of hTf guarantees anchor points to anion comprising carboxylate group available and near electron donor group (interlocking site model [44], achieving Fe^{3+} distorted octahedral coordination [45], ions like malonate, oxalate, and glycolate can easily replace synergetic carbonate, which is the most relevant synergetic anion under physiological conditions [46]. In contrast, non-synergetic anions (SCN^- , BF_4^-) attach to allosteric hTf sites, and it has been proposed [47] that it cause hTf structural changes, which can disturb Fe^{3+} binding center and leading to iron release [48]. In this scenario, the previously reported anionic carbon dots did not seem to have any anionic effect over hTf. In the opposite, the presented carbon dots act as a non-synergetic anion, where anionic CQD binds allosterically to hTf site, causing protein structural changes and weakening iron affinity to hTf until release. Iron release induced by the nanoparticles agrees with the structural changes observed in Fig. 3b, as well as with the complex force components displayed by the determined thermodynamic properties of the Fig. 3d.

4. Conclusions

Low toxic CQDs are synthesized and characterized. CQDs attached tightly via hydrophobic forces and induced structural changes to both plasma proteins (BSA and hTf). Additionally, interaction with hTf was also driven by electrostatic interaction, and the experienced structural changes lead iron to be released from hTf lobes. The uses of the nanoparticle in the biological media open questions about subtle toxicological consequences of low toxic nanoparticles. Finally, the mechanism by which iron is removed from the transferrin is especially relevant to the treatment of poisonous iron overload. Some blood disorders (sickle cell anemia and thalassemia) require a frequent whole blood transfusion, because of the lack of a human physiological mechanism to excrete iron.

Declaration of Competing Interest

None.

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